# Monopolistic Competition

EC 111 Principles of Microeconomics

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# Reading

- These slides are not a substitute for reading Mankiw, Chapter 17, Monopolistic Competition.
- Required reading: Mankiw, *Principles of Microeconomics*, 11th edition, Chapter 17, "Monopolistic Competition".

# What you must know from this chapter

1. Definition: many firms, differentiated products, free entry.
2. Why each firm faces a downward-sloping demand curve.
3. The short run: MR $=$ MC, the price from demand, profit or loss on the graph.
4. The long run: entry and exit drive profit to zero, $P =$ ATC.
5. Comparison with perfect competition: excess capacity and a markup, $P >$ MC.

Part 1

# What Is Monopolistic Competition?

# Monopolistic competition

**Definition: Monopolistic competition**

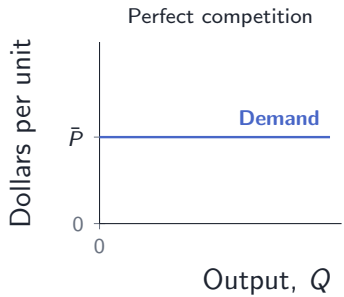
A market with many firms selling products that are similar but not identical, and free entry and exit.

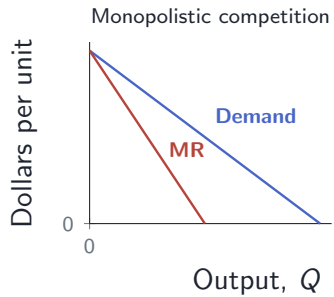
- **Many sellers:** many firms compete for the same group of buyers.
- **Product differentiation:** each firm's product is slightly different from its rivals'.
- **Free entry and exit:** firms can enter or leave the market without restriction.

**Examples:** restaurants, hair salons, coffee shops, clothing brands.

# Each firm faces a downward-sloping demand curve

- A restaurant that raises its price loses some customers to rivals.
- It does not lose them all: some prefer its food.
- So its demand curve slopes down.
- As for a monopoly, $P > \text{MR}$.



# Between perfect competition and monopoly

| | Perfect competition | Monopolistic competition | Monopoly |
|---|---|---|---|
| Number of firms | Many | Many | One |
| Product | Identical | Differentiated | No close substitutes |
| Entry | Free | Free | Blocked |
| Firm's demand | Horizontal | Slopes down | Slopes down |

**Finding**

Like perfect competition: many firms and free entry. Like monopoly: a downward-sloping demand curve, so $P >$ MR.

# Part 1 summary

- Monopolistic competition: many firms, differentiated products, free entry and exit.
- Each firm's product differs a little from its rivals'.
- So each firm faces a downward-sloping demand curve.
- So, as for a monopoly, $P >$ MR.
- Compared with perfect competition: the same many firms and free entry, but differentiated products.

# Your turn: Check your understanding

1. Which market is monopolistically competitive?
   A. Wheat farming. B. Hair salons in a city. C. A town's water network.
2. A hair salon raises its price. It loses:
   A. none of its customers. B. some of its customers. C. all of its customers.
3. New salons can open whenever they like.
   A. True B. False

# Solution: Check your understanding

1. **B.** Many salons, each slightly different, free entry. Wheat is identical (perfect competition); the water network is a natural monopoly.
2. **B.** Some customers prefer this salon, so not all leave. Demand slopes down.
3. **A.** True: entry is free.

# Check: Part 1

A monopolistically competitive industry is characterized by

A many firms, differentiated products, and barriers to entry.

B many firms, differentiated products, and free entry.

C a few firms, identical products, and free entry.

D a few firms, differentiated products, and barriers to entry.

# Check: Part 1 – answer

A monopolistically competitive industry is characterized by

A many firms, differentiated products, and barriers to entry.

**B many firms, differentiated products, and free entry.**

C a few firms, identical products, and free entry.

D a few firms, differentiated products, and barriers to entry.

**Finding**

The three features of monopolistic competition: many firms, differentiated products and free entry.

Part 2

# The Short Run: Act Like a Monopoly

# The same three steps as a monopoly

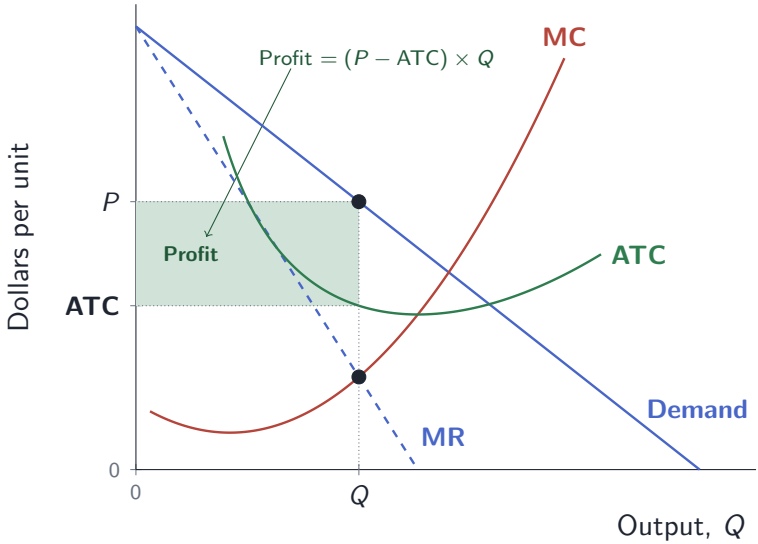
1. **Output:** produce $Q$ where MR $=$ MC.
2. **Price:** go up from $Q$ to the demand curve. That is the price, $P$.
3. **Profit:** compare $P$ with ATC at $Q$:

$$\text{Profit} = (P - \text{ATC}) \times Q.$$

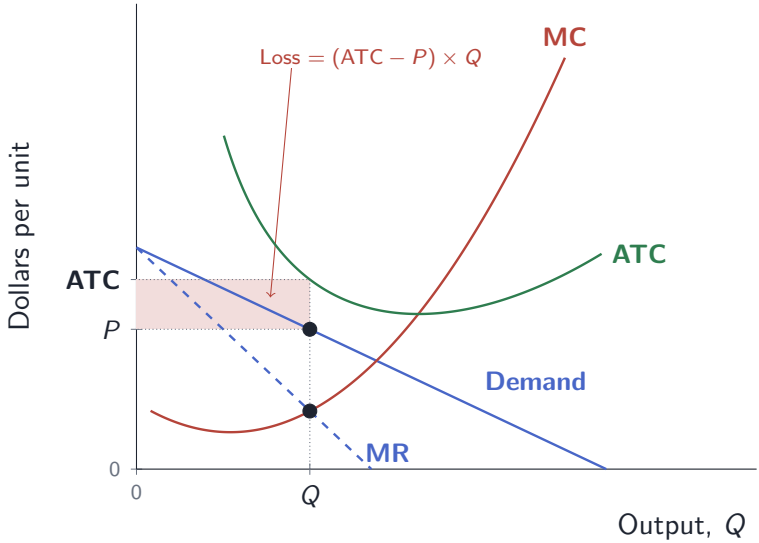
**Finding**

If $P >$ ATC, the firm makes a profit. If $P <$ ATC, it makes a loss.

# Short-run profit

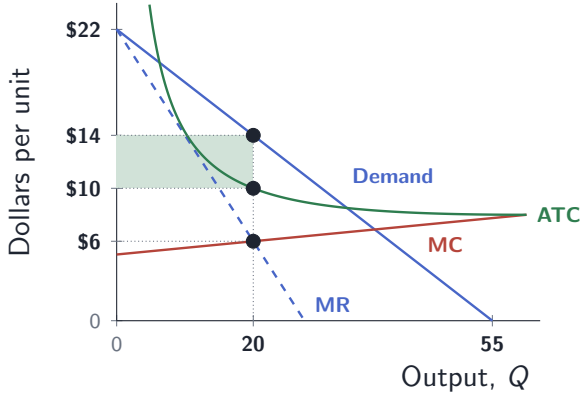


# Short-run loss



# Example: reading profit from the graph

1. MR = MC at $Q = 20$.
2. Up to demand: $P = \$14$.
3. ATC at 20 units: \$10.
4. Profit $= (14 - 10) \times 20 = \$80$.




Source of the numbers: demand $P = 22 - 0.4Q$; TC $= 90 + 5Q + Q^2/40$, so MC $= 5 + Q/20$ and ATC $= 90/Q + 5 + Q/40$.

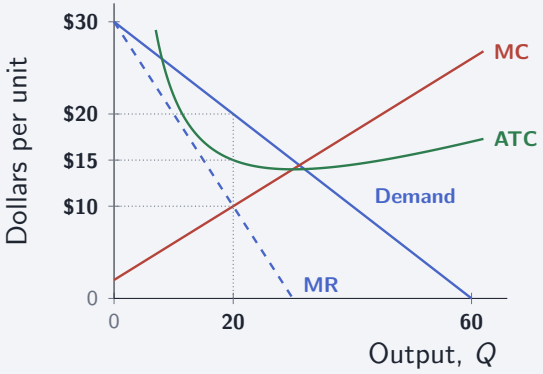
# Part 2 summary

- In the short run the firm acts like a monopoly.
- Output: where MR = MC.
- Price: read from the demand curve at that output.
- Profit = $(P - \text{ATC}) \times Q$.
- If $P < \text{ATC}$, the firm makes a loss of $(\text{ATC} - P) \times Q$.

# Your turn: Check your understanding

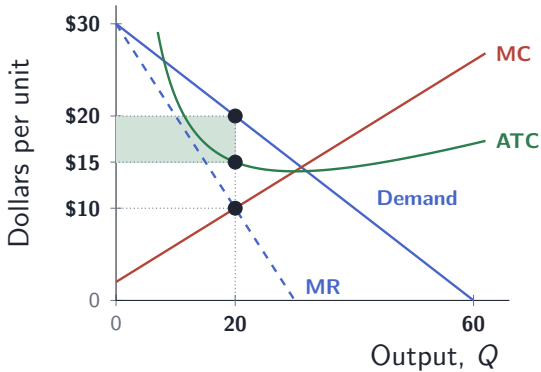
The firm maximizes profit.

1. Its output is:
   A. 10  B. 20  C. 60
2. Its price is:
   A. \$10  B. \$15  C. \$20
3. Its profit is:
   A. \$100  B. \$200  C. \$300



# Solution: Check your understanding

1. **B.** MR meets MC at 20 units (both \$10).
2. **C.** Up to demand at 20 units: \$20.
3. **A.** ATC at 20 is \$15. Profit $= (20 - 15) \times 20 = \$100$.



# Check: Part 2

In the short run, a firm operating in a monopolistically competitive market

A produces an output level where marginal revenue equals average total cost.

B produces an output where marginal revenue equals marginal cost, and the price is determined by demand.

C must earn zero economic profits.

D maximizes revenues as well as profits.

# Check: Part 2 – answer

In the short run, a firm operating in a monopolistically competitive market

A produces an output level where marginal revenue equals average total cost.

**B produces an output where marginal revenue equals marginal cost, and the price is determined by demand.**

C must earn zero economic profits.

D maximizes revenues as well as profits.

**Finding**

Steps 1 and 2: output where MR = MC, then the price from the demand curve.

Part 3

# The Long Run: Entry and Exit

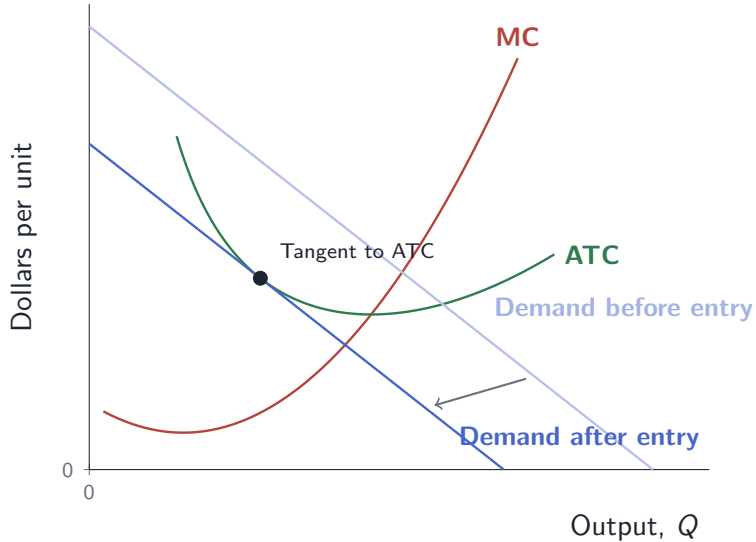
# Profit attracts entry; losses cause exit

- **Profit:** new firms enter. Buyers now have more products to choose from.
- Each existing firm loses some customers: its demand curve shifts left.
- Entry continues until profit is zero.
- **Losses:** some firms exit. Their customers move to the firms that remain.
- Each remaining firm's demand curve shifts right, until the loss is zero.

**Finding**

Entry and exit move each firm's demand curve until profit is zero.

# Entry on the graph



# Long-run equilibrium: two conditions

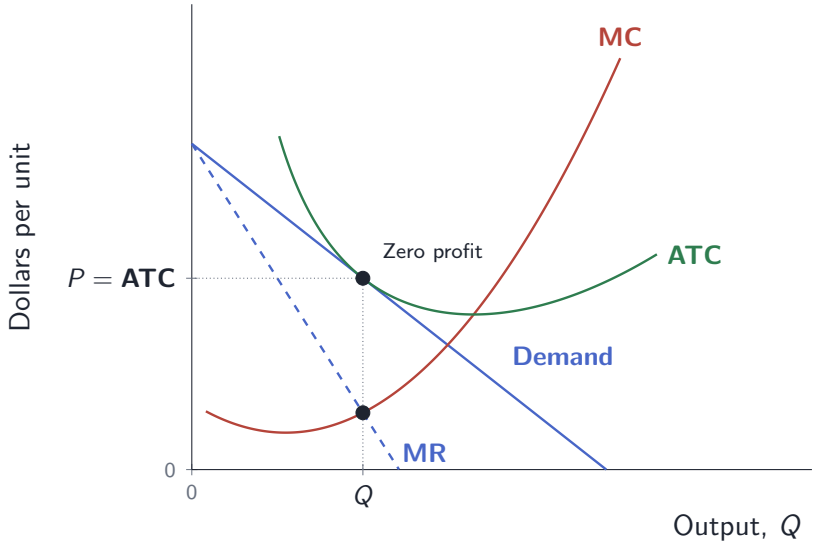
1. **The firm maximizes profit:** MR = MC.
2. **Profit is zero:** $P$ = ATC.

- Profit = $(P - \text{ATC}) \times Q = 0 \times Q = 0$.
- On the graph, the demand curve just touches the ATC curve (it is tangent to ATC).

**Finding**

In the long run, MR = MC and $P$ = ATC: demand is tangent to ATC and profit is zero.

# Long-run equilibrium on the graph



# Part 3 summary

- Profit attracts entry: each firm's demand shifts left.
- Losses cause exit: each remaining firm's demand shifts right.
- This stops when profit is zero.
- Long run: MR $=$ MC and $P =$ ATC.
- On the graph, demand is tangent to ATC.
- Compared with perfect competition: long-run profit is zero in both.

# Your turn: Check your understanding

Coffee shops in a city are making a profit.

1. New shops open. Each existing shop's demand curve:
   A. shifts left. B. shifts right. C. does not move.
2. In the long run, each shop's profit is:
   A. positive. B. zero. C. negative.
3. In the long run, a shop sells 300 cups a day at \$4 a cup. Its ATC is:
   A. \$3. B. \$4. C. \$5.

# Solution: Check your understanding

1. **A.** Buyers spread over more shops, so each shop sells less at every price.
2. **B.** Entry continues until profit is zero.
3. **B.** Zero profit means $P = \text{ATC}$, so ATC $= \$4$.

# Check: Part 3

In monopolistically competitive markets, free entry and exit suggests that

A the market structure will eventually be characterized by perfect competition in the long run.

B all firms earn zero economic profits in the long run.

C some firms will be able to earn economic profits in the long run.

D some firms will be forced to incur economic losses in the long run.

# Check: Part 3 – answer

In monopolistically competitive markets, free entry and exit suggests that

A the market structure will eventually be characterized by perfect competition in the long run.

**B all firms earn zero economic profits in the long run.**

C some firms will be able to earn economic profits in the long run.

D some firms will be forced to incur economic losses in the long run.

**Finding**

Entry removes profit and exit removes losses, so long-run profit is zero.

Part 4

# Monopolistic Competition versus Perfect Competition

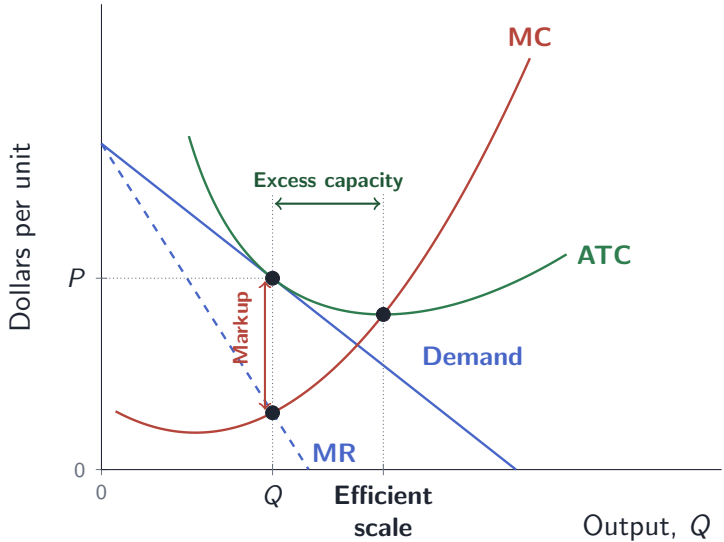
# Two differences in the long run

**Definition: Efficient scale**

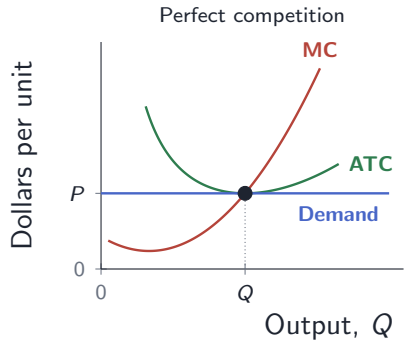
The output at which average total cost is at its minimum.

- A competitive firm produces at the efficient scale, with $P = \text{MC}$.
- **Excess capacity:** a monopolistically competitive firm produces *less* than the efficient scale.
- **Markup:** it charges a price above marginal cost, $P > \text{MC}$. The markup is $P - \text{MC}$.

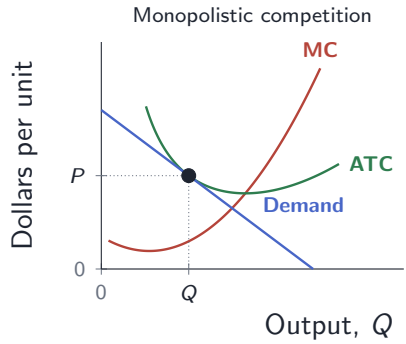
# Excess capacity and markup on the graph



# The two long runs side by side




$P = \text{MC}$. Output at the efficient scale.



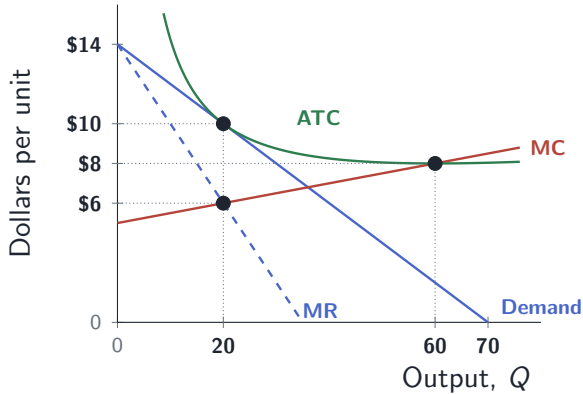

$P > \text{MC}$. Output below the efficient scale.

**Finding**

Profit is zero in both. Only perfect competition reaches the efficient scale with $P = \text{MC}$.

# Example: a firm in long-run equilibrium

1. MR = MC at $Q = 20$; MC = \$6.
2. Demand at 20: $P = \$10 =$ ATC. Profit is zero.
3. Markup $= P - \text{MC} = 10 - 6 = \$4$.
4. Efficient scale: 60 units, ATC = \$8.
5. Excess capacity $= 60 - 20 = 40$ units.




Source of the numbers: demand $P = 14 - 0.2Q$; $\text{TC} = 90 + 5Q + Q^2/40$, so $\text{MC} = 5 + Q/20$ and $\text{ATC} = 90/Q + 5 + Q/40$.

# The comparison in a table

| Long run | Perfect competition | Monopolistic competition |
|---|---|---|
| Profit | Zero | Zero |
| Price and ATC | $P = \text{ATC}$ | $P = \text{ATC}$ |
| Price and MC | $P = \text{MC}$ | $P > \text{MC}$: a markup |
| Output and efficient scale | At the efficient scale | Below it: excess capacity |

**Finding**

Because $P > \text{MC}$, some buyers who value a unit above its cost do not buy it. That is why monopolistic competition is inefficient.

# Part 4 summary

- Efficient scale: the output at minimum ATC.
- Excess capacity: the firm produces less than the efficient scale.
- Markup: the firm charges $P > \text{MC}$; the markup is $P - \text{MC}$.
- Long-run profit is zero, as in perfect competition.
- Compared with perfect competition: a competitive firm produces at the efficient scale with $P = \text{MC}$.

# Your turn: Check your understanding

A firm in long-run equilibrium sells 20 units at \$15. Its MC at 20 units is \$10. Its ATC is lowest, \$14, at 30 units.

1. The markup is:
   A. \$1  B. \$5  C. \$15
2. The excess capacity is:
   A. 0 units  B. 10 units  C. 30 units
3. A competitive firm with these costs would, in the long run, charge:
   A. \$10  B. \$14  C. \$15

# Solution: Check your understanding

1. **B.** Markup $= P - \text{MC} = 15 - 10 = \$5$.
2. **B.** Efficient scale minus output: $30 - 20 = 10$ units.
3. **B.** In the long run a competitive firm produces at the efficient scale, so $P =$ minimum ATC $= \$14$.

# Check: Part 4

In a long-run equilibrium,

A only a perfectly competitive firm operates at its efficient scale.

B only a monopolistically competitive firm operates at its efficient scale.

C neither a competitive firm nor a monopolistically competitive firm charges a markup over marginal cost.

D both a perfectly competitive firm and a monopolistically competitive firm operate at their efficient scale of production.

# Check: Part 4 – answer

In a long-run equilibrium,

**A** **only a perfectly competitive firm operates at its efficient scale.**

B only a monopolistically competitive firm operates at its efficient scale.

C neither a competitive firm nor a monopolistically competitive firm charges a markup over marginal cost.

D both a perfectly competitive firm and a monopolistically competitive firm operate at their efficient scale of production.

**Finding**

The monopolistically competitive firm has excess capacity: it produces below the efficient scale.

# Which rule answers which question

| Question | Rule |
|---|---|
| What defines the market? | Many firms, differentiated products, free entry. |
| What demand does a firm face? | Downward-sloping, so $P > \text{MR}$. |
| How much does it produce? | $\text{MR} = \text{MC}$. |
| What price does it charge? | The price on the demand curve at that output. |
| Short-run profit? | $(P - \text{ATC}) \times Q$; a loss if $P < \text{ATC}$. |
| Long-run profit? | Zero: $P = \text{ATC}$, demand tangent to ATC. |
| How does it compare with competition? | Excess capacity and a markup, $P > \text{MC}$. |

Part 5

# Advertising and Brand Names

# Advertising: the critique and the defense

Firms with differentiated products advertise. The more differentiated the product, the more they advertise: cold medicines are advertised heavily on television; industrial drill presses are not.

**The critique**

- Advertising shapes tastes rather than informing.
- It makes demand less elastic.
- So the firm can charge a larger markup, $P - \text{MC}$.

**The defense**

- Advertising tells buyers about prices, new products and new sellers.
- That makes demand more elastic and competition stronger.
- So groups of sellers sometimes lobby to restrict advertising.

# Example: advertising as a signal of quality

Two firms launch a new frozen pizza. A national advertising campaign costs \$3 million.

- **Good pizza:** buyers who try it come back. Profit from sales: \$5 million. With the campaign: $5 - 3 = +\$2$ million.
- **Poor pizza:** buyers try it once and do not return. Profit from sales: \$1 million. With the campaign: $1 - 3 = -\$2$ million.
- Only the firm with the good pizza advertises.

**Finding**

The willingness to spend heavily on advertising signals quality. What the advertisement says matters less than the fact that it is expensive.

# Brand names

Example: brand-name ibuprofen and a store brand contain the same active ingredient, but the brand costs more.

**The critique**

- Buyers pay more for a brand that is the same as the generic.
- The brand creates a difference that is not real.

**The defense**

- A brand tells buyers about quality when quality is hard to judge: a traveler picks a hotel chain in a strange town.
- The firm has a reason to keep quality high, to protect its name.